

Project Requirement Document

Project Name: E-Commerce Web Application

The system will allow users to browse products, add items to cart, place orders, and track deliveries. Admin will be able to manage products, categories, users, and orders. The platform should be scalable, secure, and mobile-responsive.

1. User Authentication & Profile

- User can register with email and password
- User can log in and log out securely
- User can reset password via email
- User can view and update their profile
- User can view their order history

2. Product Browsing & Search

- User can browse products by category
- User can search products by name and keyword
- User can filter products by price, rating, and availability
- User can view detailed product page with images and specs
- User can view related product recommendations

3. Shopping Cart

- User can add product to cart
- User can remove product from cart
- User can update quantity of items in cart
- Cart should persist across sessions
- Cart should display total price with taxes

4. Order & Checkout

- User can place an order from cart
- User can select shipping address
- User can pay via credit card, UPI, or wallet
- User receives order confirmation email
- User can track order delivery status

5. Admin Panel

- Admin can add, update, and delete products
- Admin can manage product categories
- Admin can view and update order statuses
- Admin can view registered users
- Admin can generate sales reports

6. Database & Infrastructure

- Design and maintain relational database schema
- Optimize queries for product search and cart operations
- Set up database backups and migrations
- Ensure data integrity and indexing

7. Testing & QA

- Write unit tests for all backend APIs
- Perform integration testing for cart and payment flows
- Conduct UI testing for all frontend screens
- Regression testing before each release